

No Weak Abelian Square of Total Length at Most 2745 in a Proposed Five-Letter Cyclic Morphic Word

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Abstract

Let h be the 14-uniform morphism on $\mathbb{Z}/5\mathbb{Z}$ defined by $h(0) = 01213101314310$ and $h(r) = h(0) + r$ letterwise modulo 5, and let $w = h^\omega(0)$. Shallit proposed w as a possible infinite word with no two consecutive nonempty factors having the same normalized Parikh vector. This note does not resolve that infinite problem. We prove the global finite-length result that every such pair x, y occurring consecutively anywhere in w satisfies $\Psi(x)/|x| \neq \Psi(y)/|y|$ whenever $|x| + |y| \leq 2745$. The proof combines a two-superblock embedding, valid at every starting position, with an exact exhaustive certificate for $h^4(0)$, a word of length 38,416. Consequently, any counterexample to the proposed construction has total length at least 2746.

1 Statement and context

For a finite word u on $A = \mathbb{Z}/5\mathbb{Z} = \{0, 1, 2, 3, 4\}$, write

$$\Psi(u) = (|u|_0, |u|_1, |u|_2, |u|_3, |u|_4) \in \mathbb{N}^5$$

for its Parikh vector. Two nonempty words are *weakly abelian equivalent* here when their normalized Parikh vectors agree. This notion is related to the weak abelian terminology studied by Avgustinovich and Puzynina [1]; for general background on abelian combinatorics on words, see Fici and Puzynina [2].

Put

$$b = 01213101314310, \quad h(r) = b + r \pmod{5}$$

letterwise, and extend h by concatenation. Since $h(0)$ starts with 0, the nested words $h^k(0)$ define the one-sided fixed point $w = h^\omega(0) = w_0 w_1 w_2 \dots$. Shallit proved a related construction on 16 letters, listed this five-letter morphism as a candidate, and reported that he had been unable to prove it [3, Sec. 5]. The result below does not settle that proposal.

Theorem 1. *For every $s \in \mathbb{N}_0$ and all $m, n \geq 1$ with $m + n \leq 2745$, let*

$$x = w_s \cdots w_{s+m-1}, \quad y = w_{s+m} \cdots w_{s+m+n-1}.$$

Then

$$n\Psi(x) \neq m\Psi(y).$$

Equivalently, $\Psi(x)/m \neq \Psi(y)/n$. Hence, if the proposed infinite avoidance property is false, every counterexample has $m + n \geq 2746$.

The point of the quantifier over s is that the theorem is not merely a prefix computation. Its global part is the factor embedding proved next.

2 A global two-superblock embedding

For $t \in A$ and a word u , denote by $u + t$ the word obtained by adding t to every letter modulo 5. The cyclic definition gives

$$h^k(a + t) = h^k(a) + t \quad (a, t \in A, k \geq 0). \quad (1)$$

Lemma 2. *Consecutive letters of w are distinct.*

Proof. The word b has no equal consecutive letters, and its first and last letters are both 0. If a word u has distinct consecutive letters, then the letters within each image $h(a) = b + a$ are distinct consecutively, while the boundary between $h(a)$ and $h(c)$ is the pair ac and is also distinct. Induction applies to every $h^k(0)$, and every adjacent pair of w lies in one of these prefixes. $\square$

Lemma 3. *Every factor of w of length at most 2745, together with any specified internal split, is carried by a cyclic permutation of the alphabet to a factor with the same split in $h^4(0)$.*

Proof. Set $B = 14^3 = 2744$ and use the aligned level-3 decomposition

$$w = h^3(w_0)h^3(w_1)h^3(w_2) \cdots$$

A factor of length at most $B + 1$ meets at most two level-3 blocks. Indeed, even if it begins at the last letter of one block, meeting a third block requires length $B + 2$.

If the factor lies in one block $h^3(a)$, translation by $-a$ places it in $h^3(0)$, which is the first level-3 block of $h^4(0) = h^3(b)$.

Suppose instead that it meets two consecutive blocks $h^3(a)h^3(c)$. By Lemma 2, $a \neq c$. For the nonzero difference $d = c - a \pmod{5}$, the base word b contains an adjacent pair with the same difference:

$$01 \quad (d = 1), \quad 13 \quad (d = 2), \quad 14 \quad (d = 3), \quad 21 \quad (d = 4).$$

Write this representative as uv . Translation by $u - a$, together with (1), maps

$$h^3(a)h^3(c) \quad \text{to} \quad h^3(u)h^3(v).$$

The latter pair occurs literally in $h^3(b) = h^4(0)$. The translation acts letterwise, so it preserves the factor offsets and its specified internal split. It also only permutes the five coordinates of every Parikh vector. $\square$

The endpoint $B + 1$ in Lemma 3 is important: the embedding uses a factor's total length, so it applies directly to the concatenation xy , rather than separately to x and y .

3 The exact finite certificate

Let $z = h^4(0)$ and $N = |z| = 14^4 = 38,416$. For a nonzero vector $v \in \mathbb{N}^5$, define its primitive signature

$$\text{prim}(v) = \frac{v}{\gcd(v_0, v_1, v_2, v_3, v_4)}.$$

For nonempty words u, v , their normalized Parikh vectors agree if and only if

$$\text{prim}(\Psi(u)) = \text{prim}(\Psi(v)). \quad (2)$$

This converts the finite check into integer equality only.

The supplied C++ program performs the following exhaustive sweep. For each split q with $1 \leq q < N$, it inserts

$$\text{prim}(\Psi(z_p \cdots z_{q-1})) \quad (0 \leq p < q)$$

into an open-addressed hash table, then queries the signatures of

$$\text{prim}(\Psi(z_q \cdots z_{r-1})) \quad (q < r \leq N).$$

Hashing selects a table slot only; a reported match requires equality of the complete packed signature. Each coordinate is at most $N < 2^{16}$, so the packing is lossless. Thus every triple $0 \leq p < q < r \leq N$ occurs exactly once, at its unique split q , and a completed sweep without a match proves that z has no offending pair.

The completed run examined

$$\sum_{q=1}^{N-1} q = \sum_{q=1}^{N-1} (N - q) = \frac{N(N-1)}{2} = 737,875,320$$

intervals on each side of the splits and found no equality in (2). The certificate records all $N - 1 = 38,415$ completed splits and no witness. A separate Python verifier regenerates $h^3(0)$ and $h^4(0)$, checks the completion totals, and verifies every nonzero cyclic difference representative used in Lemma 3. These checks validate the stated finite computation; they do not by themselves extend it to the infinite word.

Exact finite certificates based on exhaustive integer computation also arise in other finite-library problems in discrete geometry; for one methodologically related example whose scope is explicitly finite, see Zeng et al. [4].

Proof of Theorem 1. Fix s, m, n as in the theorem and regard the concatenation xy as one factor of length $m + n \leq 2745$ with a split after its first m letters. Lemma 3 maps this factor and split, up to a common permutation of Parikh coordinates, into $h^4(0)$. An equality $n\Psi(x) = m\Psi(y)$ would therefore give an equality of primitive signatures at one of the exhaustively checked splits, contrary to the finite certificate. The final lower bound is the integer contrapositive. $\square$

4 Scope and reproducibility

The argument controls exactly the range $m + n \leq 2745$. No desubstitution or alignment reduction is presently supplied for larger factors, so neither the theorem nor the computation decides whether w avoids weak abelian squares at every length. The contribution relative to the proposed morphism in [3] is the all-starting-position lower bound and its reproducible finite certificate, not a resolution or a priority claim.

The supplementary files needed to reproduce the computation are

```
evidence/exhaustive_h4.cpp,
evidence/h4_collinearity.json, and
evidence/verify_h4_certificate.py.
```

A representative build and run is

```
clang++ -std=c++20 -O3 -DNDEBUG -march=native \
  evidence/exhaustive_h4.cpp -o /tmp/exhaustive_h4
/tmp/exhaustive_h4
/Users/mac/4prove-or-disprove-math/.research-venv/bin/python \
  evidence/verify_h4_certificate.py
```

The C++ sweep uses exact integer counts and equality tests; reported elapsed time is not part of the certificate.

Computational and AI assistance. Exact exhaustive computation was used only for the finite word $h^4(0)$, with a separate Python verifier for word generation, totals, and block coverage. An OpenAI language model assisted with organizing the proof, inspecting the computational artifacts, and drafting the exposition. A separate model-context audit checked the mathematical scope and exact certificate; this was not external peer review. All claims in the theorem are supported by the displayed argument and the named exact artifacts.

References

- [1] Sergey V. Avgustinovich and Svetlana Puzynina. Weak abelian periodicity of infinite words. *Theory of Computing Systems*, 59(2):161–179, 2016. <https://doi.org/10.1007/s00224-015-9629-1>.
- [2] Gabriele Fici and Svetlana Puzynina. Abelian combinatorics on words: A survey. *Computer Science Review*, 47:100532, 2023. <https://doi.org/10.1016/j.cosrev.2022.100532>.
- [3] Jeffrey Shallit. An infinite walk in $\mathbb{N}^{16}$, using only unit steps, with no three collinear points, 2026. Version 1, 5 September 2026, <https://arxiv.org/abs/2609.05780>.
- [4] Zijian Zeng, Houde Liu, Kuru Ratnavelu, and Ong Seng Huat. Exact finite certificates for the four-dimensional ternary borsuk problem and five-dimensional ternary kissing codes. *Preprints*, August 2026. <https://doi.org/10.20944/preprints202608.1658.v1>.